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Buy-side divergence of opinion and stock returns: Evidence from call auctions

Yuntian Zhang [a], Yongjie Zhang [a,b,*], Zhenao Guo [a]

[a] College of Management and Economics, Tianjin University, China
[b] Laboratory of Computation and Analytics of Complex Management Systems (CACMS), Tianjin University, China

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ABSTRACT

This paper uses the call auctions of the Chinese stock market to provide high-frequency evidence on how divergence of opinion under short-sale frictions is reflected in prices. By leveraging the institutional design of the call auction – specifically the requirement to commit non-cancellable orders prior to the determination of the opening price – we isolate the formation of opinion divergence from the price determination process in a temporal sequence. We construct a buy-side divergence measure from order-level bid prices submitted during the opening call auction and find that stocks with higher buy-side divergence earn significantly higher overnight returns at the open (3.05% per month for the long-short portfolio). This rise in the opening price largely reverses in the continuous trading session of the same day, resulting in an intraday overvaluation–correction pattern consistent with the hypothesis of Miller (1977). These effects are amplified by tighter short-sale constraints, corroborating that frictions hinder the incorporation of pessimistic opinions into prices. Furthermore, mechanism tests favor the "constrained pessimists" channel over alternative explanations, such as "absent pessimists" or mechanical price pressure. By employing a novel measure and identification strategy, this study helps fill a gap regarding the contemporaneous price impact of divergence and provides granular evidence on its microstructure foundations.

1. Introduction

A large empirical literature has shown that under short-sale constraints, opinion divergence is positively related to contemporaneous returns but negatively related to subsequent returns, consistent with the hypothesis of Miller (1977) that optimistic beliefs are capitalized into prices and later corrected. However, because divergence and same-period returns are highly endogenous, and suitable high-frequency data and market settings are rare, most existing studies focus on the predictive relation between divergence and future returns. As a result, we still lack direct evidence on whether divergence itself plays an active role in bidding up prices within the same trading period.

This paper exploits the call auction mechanism and the order-level buy-side limit orders in the Chinese stock market to examine how investor divergence of opinion is related to contemporaneous returns. Between 9:15 and 9:25 each trading day, investors can submit orders but no trades are executed; at 9:25, the trading system matches all orders at once according to a maximum-volume rule and sets the opening price. During this window, buy-side divergence can be measured directly from the dispersion of order quotes,[1] and is largely insulated from immediate price feedback and liquidity-motivated trading. The overnight return can therefore

* Corresponding author.
E-mail addresses: zytcomer@tju.edu.cn (Y. Zhang), yjz@tju.edu.cn (Y. Zhang), guozhenao@tju.edu.cn (Z. Guo).

[1] Garfinkel (2009) argues that divergence of opinion extracted from order data provides a better proxy than other measures.

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be viewed as a relatively direct economic outcome of divergence formed during the call auction. Compared with previous studies that mainly rely on daily, weekly or monthly data, this setting substantially mitigates concerns about reverse causality between divergence and contemporaneous returns and provides cleaner empirical evidence on the mechanism proposed by Miller (1977).

Using order-level buy quotes from the opening call auction,[2] we construct a volume-weighted standard deviation of order prices as a measure of buy-side opinion divergence and obtain four main findings. First, higher buy-side divergence in the opening call auction is associated with higher overnight returns: a high-minus-low divergence portfolio earns an overnight return spread equivalent to about 3.05% per month, but this spread fully reverses during the same day's intraday session, with an intraday spread of about −3.48% per month, a pattern consistent with same-day "overvaluation–correction" in the spirit of Miller (1977). Second, both the overnight return and the intraday reversal are markedly stronger for stocks facing tighter short-sale constraints, supporting the "constrained pessimist" channel. Third, our results remain robust after accounting for alternative explanations, such as the potential absence of pessimistic investors and mechanical price pressure. Finally, a similar pattern emerges in closing call auctions, where high divergence predicts closing returns that reverse the following day, reinforcing the paper's core conclusions.

Our study makes two primary contributions. First, at the measurement level, we introduce a granular measure of opinion divergence derived from order-level submissions during the call auction. Unlike conventional proxies such as analyst forecast dispersion or aggregate turnover,[3] which primarily reflect expert expectations or broad trading activity, our measure directly captures the distribution of subjective beliefs among actual traders. This alignment with the theoretical construct of Miller (1977) allows for a more precise identification of cross-sectional variation in disagreement. Second, at the identification level, we exploit the unique institutional setting of the Chinese call auction, where divergence is formed before clearing prices are determined. This temporal decoupling effectively mitigates endogeneity concerns regarding reverse causality between contemporaneous returns and divergence. More importantly, it provides direct empirical evidence for Miller (1977)'s core assertion that divergence bids up contemporaneous prices—a link that prior literature has struggled to isolate due to data frequency constraints.

2. Theoretical framework and hypotheses development

According to Miller (1977), when short-sale constraints are present, the opinions of pessimistic investors cannot be fully expressed through trading behavior. Consequently, stock prices are primarily dominated by the beliefs of optimistic investors, which can lead to irrational overvaluation and positive short-term returns. As prices eventually revert toward their fundamental values, a subsequent return reversal is expected. Since this theory was proposed, scholars have attempted to test it using various proxies for opinion divergence, such as analyst forecast dispersion (Boehme et al., 2006; Diether et al., 2002; Fischer et al., 2022; Kim and Ryu, 2025; Jin et al., 2025), return volatility (Berkman et al., 2009; D'Augusta et al., 2023), and trading volume (Y. Han et al., 2022; Chang et al., 2022). However, constrained by data frequency, these proxies are mostly constructed from monthly or lower-frequency data. Consequently, existing research focuses on the predictive relation between divergence and future returns over long holding periods, providing little direct evidence of how opinion divergence contemporaneously drives prices upward.

The key to overcoming this identification challenge lies in finding a market environment with a clear temporal separation between the formation of opinion divergence and the determination of prices. The call auction mechanism in the Chinese stock market provides such a unique research opportunity. Unlike mature markets in the West where the call auction is primarily driven by informed investors such as institutions,[4] the Chinese call auction is characterized by several distinct institutional features: (1) participation is effectively mandatory for all stocks at the open and close, with no alternative trading mechanisms (Q. Han et al., 2022); (2) the main board operates without market makers (designated dealers), making prices entirely order-driven (Gerace et al., 2015); and (3) the market is dominated by retail investors, ensuring the breadth of different opinions (Carpenter et al., 2021; Tan et al., 2023). These features help mitigate the interference of information asymmetry – a common confounding factor in call auction studies within developed markets – on the observed effects of opinion divergence.

Specifically, during the call auction, investors can submit orders but cannot observe real-time transaction prices.[5] Furthermore, once the market enters the non-cancellable window, submitted orders are final. This design ensures that the dispersion of bid prices largely reflects heterogeneity in investor beliefs, isolated from feedback from realized price. At the end of the auction, the system clears all orders at once based on the maximum-volume rule to set the opening price, institutionally decoupling the process of

[2] Our analysis focuses on buy orders rather than sell orders for two main reasons. First, data availability imposes constraints on the use of sell-side records. Second, sell orders in the call auction are also likely to embed liquidity and rebalancing motives, which makes it harder to interpret their dispersion as pure belief heterogeneity. By contrast, in a market with stringent short-sale constraints like China, "observable optimism" is naturally expressed through aggressive buying rather than shorting. Our research question is precisely rooted in the (Miller, 1977) framework: under short-sale constraints, prices are disproportionately shaped by the most optimistic investors. Concentrating on buy-side divergence therefore aligns our empirical design with the underlying theory and ensures that our divergence measure captures the variation in optimistic beliefs that is most relevant for contemporaneous pricing.

[3] For example, Fischer et al. (2022), Kim and Ryu (2025) and Jin et al. (2025) proxy opinion divergence using analyst forecast dispersion, while Chang et al. (2022), Y. Han et al. (2022) and D'Augusta et al. (2023) employ stock return volatility and trading volume. These measures are typically constructed from low-frequency data, limiting their ability to capture high-frequency disagreement dynamics.

[4] For example, Ellul et al. (2005) find that traders at the London Stock Exchange (LSE) tend to avoid the call auction and instead use market maker services when uncertainty is high, and Ibikunle (2015) indicates that the LSE opening auction is primarily driven by informed traders. In addition, Nasdaq and the NYSE use a hybrid system where the opening or closing price is determined jointly by auction or designated market makers (Madhavan and Panchapagesan, 2000; Cao et al., 2000). These evidence suggest that in mature markets, the call auction can be characterized by a mix of information asymmetry and opinion divergence, making it difficult to identify these disagreements.

[5] While the semi-transparent pre-opening disclosure mechanism allows investors to monitor indicative prices, it is important to note that these are merely indicative and differ from the final clearing prices determined at the end of the auction.

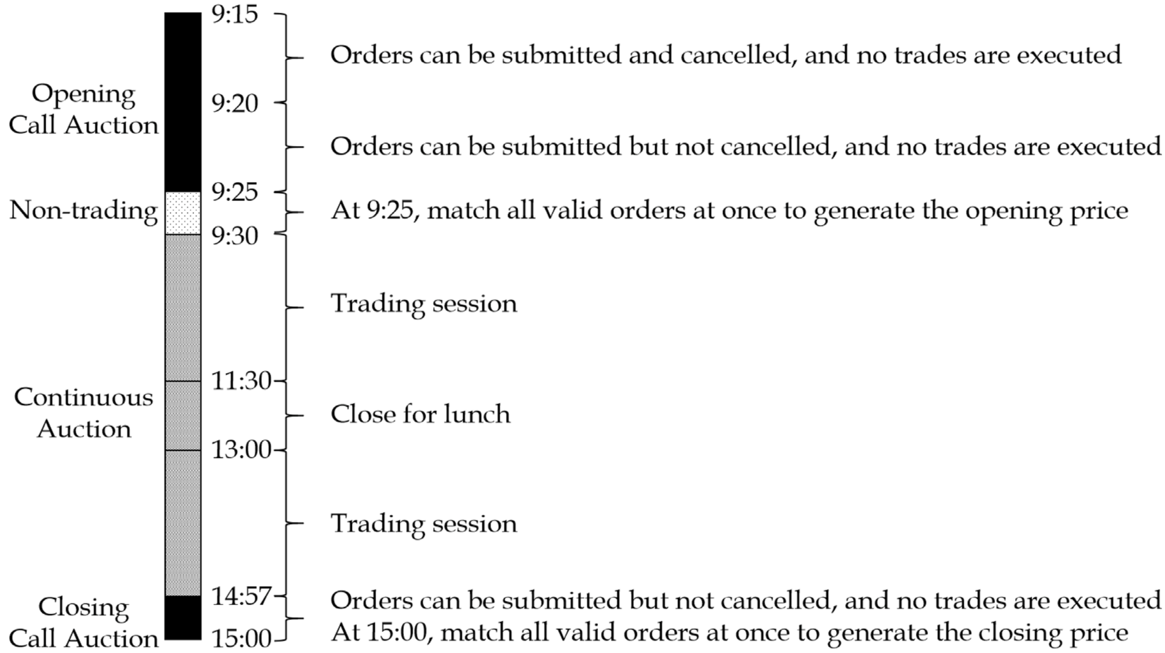

Fig. 1. Daily trading schedule and timeline of the Chinese stock market.
This figure illustrates the daily trading schedule and timeline of the Chinese stock market. Absent suspensions or other special circumstances, all stocks follow these trading rules.

opinion formation from price determination. Based on this institutional analysis and the Miller (1977) framework, we propose our first hypothesis:

$H1$: **Stocks with higher buy-side opinion divergence during the opening call auction exhibit higher overnight returns at the open.**

Once the call auction ends and continuous trading begins, all investors can observe whether the opening price aligns with their expectations. For stocks where the opening price is inflated due to high opinion divergence, the deviation from fundamentals makes a subsequent price correction more likely. Accordingly, we propose our second hypothesis:

$H2$: **Stocks with higher buy-side opinion divergence during the opening call auction exhibit lower subsequent intraday returns.**

Finally, the core premise of Miller's theory is the existence of short-sale constraints. The belief-driven price run-up can only persist if pessimists are unable to express their views through short-selling. Therefore, in stocks where short-selling is permitted, the upward pressure exerted by opinion divergence on prices should be attenuated. Based on this logic, we propose our third hypothesis:

$H3$: **The divergence of opinion effects described in $H1$ and $H2$ are significantly stronger for stocks that cannot be sold short.**

3. Institutional background, data and measurements

3.1. Institutional background: The call auction in Chinese stock market

Since 2006, the Shenzhen Stock Exchange has used a hybrid mechanism combining opening and closing call auctions with intraday continuous trading. Fig. 1 illustrates the trading timeline of a typical day. The opening call auction runs from 9:15 to 9:25: between 9:15 and 9:20 investors may submit and cancel orders, whereas from 9:20 to 9:25 they may only submit. Throughout this period the system merely accumulates orders, and no trades or transaction prices are generated. At 9:25, the book is uncrossed once and the opening price is set by a maximum-volume rule, followed by a non-trading phase from 9:25 to 9:30. Continuous trading then takes place from 9:30 to 11:30 and 13:00 to 14:57. In the final three minutes (14:57–15:00), the market re-enters a call auction in which orders can be submitted but not canceled; at 15:00, a single uncrossing again determines the closing price using the maximum-volume rule.

As discussed in the introduction, these institutional features help alleviate, to some extent, concerns about reverse causality – where returns feed back into divergence – and about confounding liquidity trading. As a result, our divergence measure is more tightly linked to the opening and closing prices.

3.2. Measuring buy-side divergence: Based on order-level buy-side orders in the call auction

The buy orders data during the call auctions are provided by a Chinese brokerage and include all stocks listed on the main board of the Shenzhen Stock Exchange (tickers starting with "0").[6] The sample spans January 4, 2010 to December 24, 2024 and covers 1582 stocks. Each order record contains the stock code, time stamp, price, and volume. To ensure data quality, we: (1) drop orders with prices outside the daily price limits (±10%) and those with volume not in multiples of 100 shares[7]; (2) select orders submitted between 9:20 and 9:25, when cancellations are not allowed, to better capture normal trading intentions; and (3) require at least ten valid orders per stock in this interval to improve the reliability of our divergence measure.

To reflect the greater influence of large volume orders in price formation, we compute a volume-weighted standard deviation of order prices and use it as our proxy for buy-side opinion divergence. The measure is constructed as follows:

$$\mu_{i,t} = \frac{\sum_j v_{i,j,t} \cdot p_{i,j,t}}{\sum_j v_{i,j,t}} \tag{1}$$

$$\sigma_{i,t} = \sqrt{\frac{\sum_j v_{i,j,t} \cdot (p_{i,j,t} - \mu_{i,t})^2}{\sum_j v_{i,j,t}}} \tag{2}$$

$$Divergence_{i,t} = \frac{\sigma_{i,t}}{P^{Close}_{i,t-1}} \tag{3}$$

where $p_{i,j,t}$ and $v_{i,j,t}$ are the price and volume of the jth buy order for stock i on day t, thus the $\mu_{i,t}$ is the average price per share. $\sigma_{i,t}$ is the volume-weighted standard deviation of order prices.[8] To enhance cross-sectional comparability, we divide the above volume-weighted standard deviation by the previous day's closing price ($P^{Close}_{i,t-1}$) to obtain a standardized divergence measure ($Divergence_{i,t}$). This normalization removes the mechanical effect of the price level on the dispersion of order prices and allows divergence across stocks to be compared on a common scale.

3.3. Other data and measurements

We obtain daily trading and accounting data, as well as announcement dates of major corporate events from the CSMAR database. These include daily returns (R), overnight returns($R^{overnight}$), intraday returns ($R^{intraday}$), the log of market capitalization ($LogME$), ROE, relative effective spreads (RES), realized volatility (RV), industry overnight return ($R^{overnight}_{industry}$), industry intraday return ($R^{intraday}_{industry}$), the natural logarithm of one plus the total number of buy orders ($OrderNumber$), the total monetary value of buy orders normalized by the stock's market capitalization ($BuyPressure$), and three categories of major events: financial reports, M&A, and violations. Minute-by-minute return data are provided by the aforementioned brokerage. Detailed formulas and variable definitions are reported in the Appendix.

In line with Liu et al. (2019), we sequentially exclude: (1) stocks with a listing history of less than six months; (2) stocks with fewer than 15 trading days in a given month; (3) stocks with fewer than 120 trading days over the past year; and (4) ST firms. The results of descriptive statistics and Pearson correlations for the main variables are presented in the Appendix Table I.

4. Empirical analysis

4.1. Empirical strategies

We conduct our empirical analysis from both portfolio and panel-regression perspectives. First, for each trading day before the opening price is set (i.e., before 9:25), we sort all sample stocks into five portfolios based on their call auction (9:20–9:25) buy-side divergence of investor opinion, from low to high. Within each quintile, we form both value-weighted and equal-weighted portfolios using float-adjusted market capitalization, and compute their overnight returns to the open and subsequent intraday returns. We then average these portfolio returns over time.

In addition, we also implement a double-sort to control for variables that may be correlated with divergence. Specifically, following (He et al., 2026), stocks are sorted into quintiles based on a control variable. Within each quintile, stocks are then further sorted into quintiles based on divergence of investor opinion. The overnight and intraday returns of the five divergence portfolios are averaged across the five control variable quintiles.

[6] The data includes all orders of the entire market, not only from orders submitted by this brokerage's clients.

[7] In the Chinese stock market, stocks with tickers starting with "0" are subject to a daily price limit of ±10%. In addition, stock trades must be in multiples of 100 shares, which is the minimum trading unit.

[8] The key idea is to conceptually split each order into multiple "unit orders" according to its volume size (treating each share as one unit), and then compute the standard deviation across the prices of all unit orders, which naturally yields a volume-weighted standard deviation. For example, an order to buy 100 shares at 10 yuan and another to buy 200 shares at 15 yuan can be viewed as 100 unit orders at 10 yuan and 200 unit orders at 15 yuan. The standard deviation computed over these 300 price observations is exactly the volume-weighted standard deviation of order prices.

Table 1
Portfolio analysis results.

Portfolios	Overnight return		Intraday return	
	VW	EW	VW	EW
Low Divergence	−3.64	−4.59	4.67	6.14
	(−9.42)	(−11.27)	(8.68)	(10.35)
2	−2.93	−4.05	3.86	5.22
	(−8.73)	(−11.64)	(7.34)	(8.85)
3	−2.36	−3.45	3.34	4.60
	(−7.68)	(−10.95)	(6.42)	(7.90)
4	−1.75	−2.56	2.61	4.03
	(−6.41)	(−9.07)	(4.91)	(7.02)
High Divergence	−0.59	−0.63	1.19	2.67
	(−2.22)	(−2.24)	(2.32)	(4.72)
High - Low	3.05	3.96	−3.48	−3.46
	(9.13)	(11.27)	(−10.46)	(−12.47)

This table reports the raw returns of portfolios sorted on buy-side divergence of opinion. Specifically, before the opening price is set at 9:25 each trading day, all sample stocks are sorted into five groups based on their divergence during the 9:20–9:25 call auction. Within each group, we form value-weighted and equal-weighted portfolios, and compute overnight (close-to-open) and intraday (open-to-close) returns. All returns are adjusted to a monthly frequency (daily returns × 21) and are expressed in percentages. The sample period is from January 4, 2010 to December 24, 2024. Newey–West adjusted t-statistics are reported in parentheses.

Second, we estimate panel regressions with both stock fixed effects and trading-day fixed effects, and include two groups of control variables capturing stock characteristics and major corporate events (please see the Appendix for detailed variable definitions):

$$R^{z}_{i,t} = \alpha + \beta_1 \cdot Divergence_{i,t} + \lambda_{\mathbf{1}} \cdot \boldsymbol{X}^{\boldsymbol{char}} + \lambda_{\mathbf{2}} \cdot \boldsymbol{X}^{\boldsymbol{events}} + \gamma_i + \sigma_t + \epsilon_{i,t} \tag{4}$$

where $R^{z}_{i,t}$ is the overnight or intraday return for stock i on day t.

4.2. Empirical results

4.2.1. Baseline results

Table 1 reports the monthly-adjusted portfolio returns, providing the first direct test of our core hypotheses. All portfolios exhibit negative overnight returns at the open, consistent with the summary statistics and prior evidence of overnight discounts under the $T+1$ trading rule (Bian et al., 2022; Qiao and Dam, 2020). More importantly, we observe a pronounced and monotonic spread across portfolios sorted by buy-side divergence. Specifically, the high-divergence portfolio earns an average overnight return of −0.59%, whereas the low-divergence counterpart earns −3.64%. The resulting long-short spread of 3.05% per month is both statistically and economically significant (t-stat = 9.13), directly supporting Hypothesis 1 (H1) that higher buy-side divergence in the call auction is associated with higher contemporaneous returns.[9]

The intraday evidence further reinforces the "overvaluation-correction" mechanism. Fig. 2 illustrates that the opening price premium is transitory: once continuous trading begins at 9:30, the return spread between the high- and low-divergence portfolios narrows almost linearly. As reported in Table 1, the high-divergence portfolio yields an intraday return of 1.19%, significantly underperforming the low-divergence portfolio's 4.67%. This yields a significant intraday spread of −3.48% (t-stat = −10.46), which effectively offsets the overnight advantage. This complete reversal provides strong empirical evidence for Hypothesis 2 (H2). Collectively, these findings document a same-day return profile that closely aligns with the theoretical predictions of Miller (1977) regarding divergence-driven mispricing and its subsequent correction. Although these patterns rely on unique order-level data that may be difficult for practitioners to exploit in real-time, they offer critical insights into the microstructure of price formation during the call auction.

To further ensure that call auction divergence effect is not simply driven by known variables such as volatility, liquidity, industry returns, or related characteristics, we conduct a series of double-sorting portfolio analyses to control for these potential confounds one by one. Columns (1) to (6) of Table 2 show that, after controlling for lagged returns, firm size, liquidity, volatility and industry returns, the high-divergence portfolio still earns significantly higher overnight returns than the low-divergence portfolio (with spreads between 2.99% and 3.98%), and that this opening premium is almost fully reversed over the rest of the trading day.

A potential concern is that aggressive order imbalances may drive our results: extreme buy quotes could mechanically inflate the divergence measure while triggering a post-open price correction. To address this, we incorporate $OrderNumber$ and $BuyPressure$[10]

[9] It is important to note that, within the context of the characteristic overnight discount in the Chinese stock market, the high-low spread reflects a relative attenuation of the negative overnight drift effect, rather than a positive overnight return.

[10] $OrderNumber$ is defined as the natural logarithm of one plus the total number of buy orders submitted between 9:20 and 9:25; $BuyPressure$ is defined as the total monetary value of these buy orders normalized by the stock's market capitalization on the previous day.

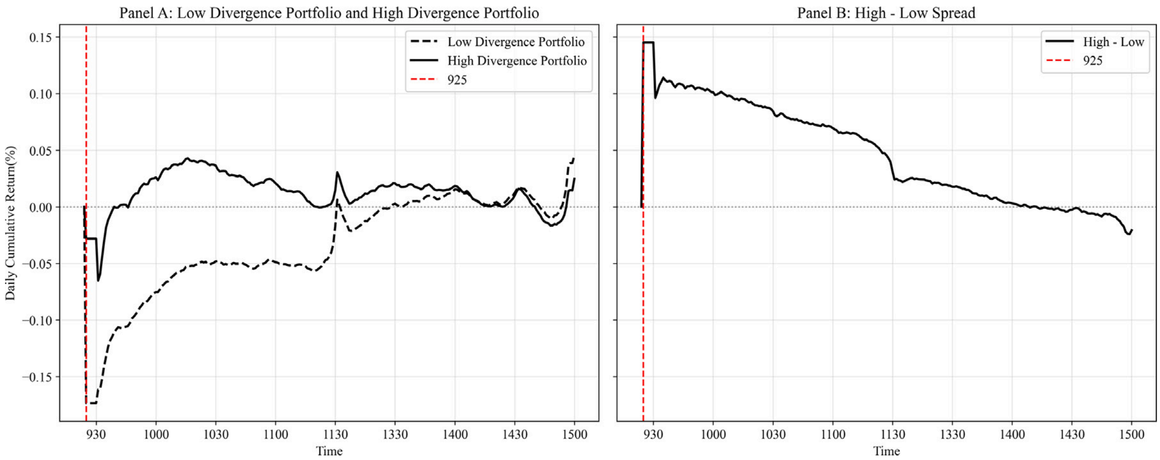

Fig. 2. Daily cumulative return of value-weighted portfolios.
This figure plots the open-to-close cumulative returns of portfolios formed on call auction opinion divergence. Specifically, before the opening price is set at 9:25 each trading day, all sample stocks are sorted into five groups based on their divergence during the 9:20–9:25 call auction. Within each group, we form value-weighted portfolios and compute minute-by-minute returns for the low-divergence, high-divergence, and high-minus-low (H–L) portfolios, which are then averaged over time. The red dashed line marks the time (9:25) at which the opening price is determined. The sample period is from January 4, 2010 to December 24, 2024.

as first-stage sorting variables in a double-sorting procedure to examine whether the divergence effect persists after neutralizing order imbalance.

The results in the final columns of Table 2 confirm that the divergence long–short spreads remain robust. For instance, after controlling for $BuyPressure$ (Column 8), the high-divergence portfolio still earns 2.34% per month higher overnight returns than the low-divergence portfolio, with this premium reversing by 2.86% intraday. While slightly attenuated relative to the baseline, these results confirm that opinion divergence is not merely a proxy for aggressive bidding behavior.

4.2.2. Panel regression

Beyond the portfolio-level evidence, we estimate panel regressions to further verify the robustness of our results, as reported in Table 3. The regression findings are highly consistent with the portfolio analysis in terms of both direction and economic magnitude. Specifically, in the baseline specification without control variables (Column 1), the coefficient on $Divergence$ is 0.0552. This suggests that a one-standard-deviation increase in Divergence (0.92%) corresponds to a 5.1 bps increase in overnight returns ($0.0552 \times 0.92\% \approx 0.051\%$). To put this into perspective, the spread between the high- and low-divergence quintiles is 2.41% (approximately 2.62 standard deviations), which implies a predicted overnight return spread of roughly 13.4 bps ($5.1bps \times 2.62$). This estimation aligns closely with the 14.5 bps long-short spread observed in Panel A of Table 1.[11] Consistent results are also obtained for intraday returns (Column 5).

Crucially, after incorporating a comprehensive set of control variable, the coefficient on $Divergence$ remains both statistically and economically significant. For macroeconomic information, these are already included in the fixed effects. By mitigating the influence of these potential confounding factors, the regression analysis provides robust evidence that our divergence measure effectively captures investor opinion heterogeneity rather than mere market noise, thereby reinforcing our primary conclusions.

5. Identification and mechanism

While our baseline results align with (Miller, 1977), the use of buy-side data necessitates distinguishing the "constrained pessimist" channel from two alternatives: the absence of pessimists or mechanical price pressure. The former suggests a scenario where pessimistic investors are simply absent from the market, which implies a lack of opinion divergence; the latter attributes the intraday reversal to mean reversion following order imbalances. This section employs a series of tests to isolate the opinion-divergence effect from these confounding factors.

[11] The average divergence values for the five quintile portfolios (0.95%, 1.41%, 1.82%, 2.32%, and 3.36% from the lowest to the highest, respectively) are not reported in the table for brevity. The resulting spread between the extreme quintiles is 2.41%. In addition, the results in Table 1 are reported on a monthly-adjusted basis. In daily terms, this corresponds to a return of 0.145% ($3.05\%/21$), approximately 14.5 basis points.

Table 2
Double-sorting.

Panel A: Overnight return								
Portfolios	(1) R_{t-1}	(2) ME_{t-1}	(3) ROE_{t-1}	(4) RES_{t-1}	(5) RV_{t-1}	(6) $R^{overnight}_{industry,t}$	(7) $OrderNumber_t$	(8) $BuyPressure_t$
Low Divergence	−3.70	−4.60	−3.89	−3.79	−4.40	−3.54	−3.73	−3.08
	(−11.45)	(−13.15)	(−11.34)	(−12.03)	(−13.52)	(−10.60)	(−11.66)	(−9.32)
2	−3.11	−4.21	−3.25	−3.40	−3.42	−2.75	−2.80	−2.53
	(−10.47)	(−13.90)	(−10.87)	(−12.17)	(−11.81)	(−9.51)	(−9.99)	(−8.42)
3	−2.39	−3.71	−2.70	−2.77	−2.60	−2.18	−2.29	−2.16
	(−8.59)	(−13.16)	(−9.74)	(−10.55)	(−9.53)	(−8.07)	(−8.93)	(−7.51)
4	−1.82	−2.80	−2.07	−2.17	−1.77	−1.51	−1.72	−1.88
	(−6.99)	(−10.66)	(−8.04)	(−8.69)	(−6.84)	(−5.99)	(−7.27)	(−6.84)
High Divergence	−0.71	−0.69	−0.70	−0.72	−0.41	−0.23	−0.40	−0.74
	(−2.83)	(−2.69)	(−2.74)	(−2.88)	(−1.61)	(−0.92)	(−1.73)	(−2.69)
High - Low	2.99	3.91	3.19	3.07	3.98	3.31	3.33	2.34
	(14.11)	(15.61)	(12.81)	(13.40)	(17.79)	(14.08)	(15.30)	(8.82)
Panel B: Intraday Return								
Portfolios	(1) R_{t-1}	(2) ME_{t-1}	(3) ROE_{t-1}	(4) RES_{t-1}	(5) RV_{t-1}	(6) $R^{intraday}_{industry,t}$	(7) $OrderNum_t$	(8) $BuyPressure_t$
Low Divergence	4.78	6.18	4.80	5.05	5.16	5.43	4.71	4.84
	(8.85)	(10.74)	(8.73)	(9.41)	(9.47)	(9.94)	(8.89)	(9.13)
2	4.00	5.38	3.96	4.34	4.11	4.76	3.53	4.13
	(7.28)	(9.19)	(7.17)	(8.06)	(7.54)	(8.65)	(6.53)	(7.63)
3	3.48	5.08	3.48	3.83	3.64	4.32	2.66	3.55
	(6.43)	(8.66)	(6.32)	(7.02)	(6.66)	(7.98)	(4.89)	(6.56)
4	2.73	4.31	2.69	3.27	2.71	3.52	1.75	3.18
	(4.98)	(7.38)	(4.81)	(5.98)	(5.01)	(6.48)	(3.24)	(5.83)
High Divergence	1.38	2.78	1.47	1.96	1.21	2.63	0.70	1.99
	(2.56)	(4.81)	(2.69)	(3.65)	(2.28)	(4.90)	(1.30)	(3.69)
High - Low	−3.40	−3.40	−3.33	−3.10	−3.96	−2.80	−4.01	−2.86
	(−13.75)	(−16.02)	(−13.38)	(−13.15)	(−16.82)	(−11.68)	(−18.58)	(−11.99)

This table reports the raw returns of double-sorting portfolios. Each day, stocks are sorted into quintiles based on a control variable (one of R_{t-1}, ME_{t-1}, ROE_{t-1}, RES_{t-1}, RV_{t-1}, $R^{overnight}_{industry,t}$, $R^{intraday}_{industry,t}$, $OrderNum_t$ and $BuyPressure_t$). Within each quintile, stocks are then further sorted into quintiles based on divergence of investor opinion. The returns of the five divergence portfolios are averaged across the five control variable quintiles. Within each portfolios, we form value-weighted and compute overnight (close-to-open) and intraday (open-to-close) returns. All returns are adjusted to a monthly frequency (daily returns × 21) and are expressed in percentages. The sample period is from January 4, 2010 to December 24, 2024. Newey–West adjusted t-statistics are reported in parentheses.

5.1. Identification: Constrained and absent pessimists

Miller (1977) posits that short-sale constraints prevent pessimistic views from being priced, allowing optimists to dominate. Lacking sell-side data, we must distinguish whether our findings reflect this "constrained pessimist" mechanism or an "absence of pessimists"—where buy-side divergence dictates prices simply because opposing views are missing.

We disentangle these mechanisms through two cross-sectional tests. First, we exploit China's margin-trading eligibility as an institutional split that implies systematically different short-sale frictions. Under the Miller (1977) channel, the divergence effect should be more pronounced in non-marginable stocks where short-selling is strictly prohibited. Second, we use opening auction volume as a proxy for seller participation intensity. If our findings are a byproduct of "absent pessimists", the effect should weaken or vanish in high-volume stocks where counterparty engagement is presumably higher.

We classify stocks according to margin-trading eligibility and then sort into buy-side divergence quintiles. Columns (1) of Table 4 shows that the divergence effect in non-marginable stocks is nearly twice that of marginable stocks: the monthly overnight spread is 3.59% versus 1.99%. Intraday reversals are similarly more pronounced in non-marginable stocks (−3.46% versus −2.75%). These findings validate Hypothesis 3 ($H3$) and support Miller (1977)'s prediction that short-sale constraints exacerbate price run-ups and corrections, supporting an interpretation that our divergence measure captures disagreement whose pricing impact is amplified when pessimistic trading is constrained.[12]

To address the "absent pessimists" alternative, we use opening auction volume ($Open\ Value-to-ME$) as a proxy for seller participation. Higher auction activity makes the "extreme absence of opposing participation" less plausible. While trading activity

[12] To further validate our findings, we employ firm size, liquidity, and volatility as alternative proxies for short-sale constraints (Appendix Table II). Consistent with the margin-eligibility analysis, the divergence effect is significantly more pronounced in small, illiquid, and volatile stocks, which is consistent with prior research (Chen et al., 2002; Nagel, 2005; Zhao et al., 2014). Furthermore, panel regressions incorporating interaction terms between $Divergence$ and these proxies (Appendix Table III) yield coefficients aligned with the "constrained pessimists" hypothesis. These results mitigate omitted-variable concerns and reinforce the robustness of our core conclusions.

Table 3
Panel regression for predicting overnight and intraday returns.

	$y = R^{overnight}_{i,t}$				$y = R^{intraday}_{i,t}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Divergence_{i,t}$	0.0552	0.0970	0.0972	0.0971	−0.0594	−0.0722	−0.0854	−0.0854
	(16.41)	(33.14)	(32.52)	(32.52)	(−16.28)	(−22.31)	(−29.60)	(−29.60)
$R_{i,t-1}$		0.0613	0.0555	0.0554		−0.0098	−0.0116	−0.0116
		(33.28)	(32.25)	(32.23)		(−2.65)	(−3.56)	(−3.57)
$LogME_{i,t-1}$		0.0007	0.0021	0.0021		−0.0030	−0.0042	−0.0042
		(7.45)	(23.22)	(23.10)		(−13.62)	(−21.12)	(−21.18)
$ROE_{i,t-1}$		0.0001	0.0000	0.0000		0.0000	0.0000	0.0000
		(1.66)	(1.46)	(1.46)		(−1.72)	(−1.79)	(−1.77)
$RES_{i,t-1}$		0.0072	0.0054	0.0054		−0.0003	0.0003	0.0003
		(14.09)	(11.84)	(11.86)		(−0.48)	(0.59)	(0.61)
$RV_{i,t-1}$		−0.0187	−0.0213	−0.0214		0.0067	0.0027	0.0027
		(−39.88)	(−35.73)	(−35.75)		(9.69)	(4.32)	(4.31)
$R^{industry}_{i,t}$			0.6205	0.6204			0.7408	0.7408
			(39.54)	(39.55)			(72.34)	(72.34)
$OrderNumber_{i,t}$			−0.0012	−0.0012			0.0016	0.0016
			(−17.79)	(−17.87)			(30.06)	(30.27)
$BuyPressure_{i,t}$			0.0139	0.0139			−0.0011	−0.0011
			(33.97)	(33.96)			(−4.89)	(−4.91)
$\mathbf{I}^{Earning}_{i,t-1}$				0.0001				−0.0013
				(1.66)				(−10.67)
$\mathbf{I}^{Violation}_{i,t-1}$				−0.0018				−0.0008
				(−9.72)				(−3.00)
$\mathbf{I}^{M\&A}_{i,t-1}$				0.0011				−0.0003
				(18.27)				(−2.89)
Stock FE	YES	YES	YES	YES	YES	YES	YES	YES
Trading Day FE	YES	YES	YES	YES	YES	YES	YES	YES
R^2	0.19%	4.12%	11.59%	11.62%	0.06%	0.23%	9.42%	9.43%

This table reports the estimation results for the panel regression of equation (4). The key explanatory variable is $Divergence_{i,t}$, defined as the volume-weighted standard deviation of stock i's buy-order prices during the call auction from 9:20 to 9:25 (excluding 9:25) on day t, and divided by the previous day's closing price. The dependent variables are stock i's overnight return (close-to-open) and intraday return (open-to-close) on day t. Control variables include stock i's return (R), the log (base 10) of market capitalization ($LogME$), ROE, relative effective spread (RES), realized volatility (RV) on day $t-1$, and industry return ($R^{industry}$), the natural logarithm of one plus the total number of buy orders ($OrderNumber$), the total monetary value of these buy orders normalized by the stock's market capitalization on the previous day ($BuyPressure$) and indicators for whether earnings ($\mathbf{I}^{Earning}$), violation ($\mathbf{I}^{Violation}$), or M&A ($\mathbf{I}^{M\&A}$) announcements are released on day $t-1$ (please refer to the Appendix for the definitions and construction of the control variables). Standard errors are two-dimensionally clustered, and t-values are reported in parentheses. The sample period is from January 4, 2010 to December 24, 2024.

does not uniquely identify pessimistic beliefs, it provides a useful check that our results are not driven by a near-empty opposing side.

Sorting by median $Open\ Value-to-ME$, we find that the divergence effect persists in both high- and low-activity groups (Table 4, Column 2). Monthly overnight spreads are 3.16% and 2.79%, respectively, with an insignificant 0.38% difference. These results indicate the effect is not driven by the absence of pessimists, as it remains robust even when counterparty participation trading activity is high.

Admittedly, opening trading value imperfectly proxies for pessimists' participation. Nevertheless, the "absent pessimists" story implies a clear cross-sectional prediction: the divergence–overnight return relation should attenuate in stocks with higher two-sided participation. We find no such attenuation, and the pronounced intraday reversal is difficult to reconcile with a pure "absence" mechanism, which would not naturally generate a systematic overvaluation–correction pattern within the same trading day. While data limitations preclude fully ruling out alternative interpretations, the evidence is more consistent with the "constrained pessimists" channel of Miller (1977).

5.2. Mechanism testing: Price pressure

While the documented one-day reversal aligns with Miller (1977), it may also stem from mean reversion following temporary price pressure. Aggressive bidding during the call auction could mechanically inflate both our divergence measure and opening prices, inducing a transient mispricing. We disentangle these two mechanisms by examining whether the divergence effect persists after controlling for aggressive order flow and price pressure proxies.

5.2.1. Overnight return

To isolate the divergence effect from price pressure, we neutralize $OrderNumber$ and $BuyPressure$ through double-sorting (Table 2, Columns 7–8). Post-neutralization, high-divergence portfolios continue to earn significantly higher overnight returns. These results confirm that the opinion divergence effect remains intact and persists independently of buy-side pressure intensity.

Table 4
Heterogeneity analysis.

Panel A: Overnight Return						
	(1) Margin Trading			(2) $OpenVolume-to-ME$		
	Marginable	Non-marginable	Difference	High	Low	Difference
Low Divergence	−2.46	−4.81	2.35	−3.47	−3.52	0.05
	(−7.04)	(−11.19)	(13.34)	(−7.12)	(−11.00)	(0.21)
2	−2.09	−4.29	2.21	−2.72	−2.97	0.25
	(−6.46)	(−11.70)	(15.13)	(−6.26)	(−10.59)	(1.04)
3	−1.60	−3.83	2.22	−2.17	−2.59	0.42
	(−5.55)	(−11.83)	(15.70)	(−5.44)	(−9.89)	(1.86)
4	−1.18	−3.01	1.82	−1.92	−2.01	0.09
	(−4.42)	(−10.37)	(12.87)	(−5.18)	(−8.50)	(0.37)
High Divergence	−0.47	−1.22	0.74	−0.30	−0.73	0.43
	(−1.75)	(−3.90)	(4.16)	(−0.69)	(−5.41)	(1.92)
High - Low	1.99	3.59	−1.60	3.16	2.79	0.38
	(6.82)	(8.76)	(−6.50)	(5.86)	(9.40)	(1.04)
Panel B: Intraday Return						
	(1) Margin Trading			(2) $OpenVolume-to-ME$		
	Marginable	Non-marginable	Difference	High	Low	Difference
Low Divergence	3.42	6.18	−2.76	5.00	4.57	0.43
	(6.32)	(10.23)	(−8.97)	(8.40)	(8.66)	(1.73)
2	2.84	5.42	−2.58	4.28	3.68	0.60
	(5.64)	(8.75)	(−7.69)	(7.30)	(7.00)	(2.24)
3	2.63	4.78	−2.15	3.75	3.10	0.66
	(5.02)	(7.78)	(−6.03)	(6.39)	(6.03)	(2.21)
4	1.93	4.05	−2.11	3.36	2.41	0.95
	(3.69)	(6.66)	(−5.77)	(5.91)	(4.54)	(3.00)
High Divergence	0.67	2.72	−2.04	1.93	0.93	1.00
	(1.28)	(4.52)	(−5.37)	(3.31)	(1.82)	(2.82)
High - Low	−2.75	−3.46	0.72	−3.07	−3.64	0.57
	(−7.95)	(−10.62)	(2.46)	(−7.16)	(−11.34)	(1.49)

This table presents value-weighted (VW) overnight and intraday returns for portfolios sorted by opinion divergence. In Column (1), the sample is partitioned into marginable and non-marginable stocks. In Column (2), stocks are grouped by the daily median of $Open\ Value-to-ME$ (High and Low). Within each subsample, stocks are sorted into quintiles by divergence of investor opinion. 'Difference' represents the return difference between the two subsamples. All returns are adjusted to a monthly frequency (daily returns × 21) and are expressed in percentages. The sample period is from January 4, 2010 to December 24, 2024. Newey–West adjusted t-statistics are reported in parentheses.

Notably, controlling for $BuyPressure$ narrows the overnight spread from 3.05% (Table 1) to 2.34%. This 23% attenuation suggests that while price pressure contributes to opening returns, it is not the primary driver. Similarly, $Divergence$ coefficients in panel regressions (Table 3, Columns 3–4) remain statistically significant after accounting for $OrderNumber$ and $BuyPressure$. In sum, these findings confirm that the opinion-divergence effect persists independently of aggressive bidding and mechanical price pressure.

5.2.2. Intraday return

To distinguish opinion divergence from price pressure in intraday trading, we perform a conditional analysis based on overnight returns. If the reversal is driven solely by opening price pressure, it should be negligible in stocks with low opening run-ups. Conversely, if rooted in opinion divergence, a significant intraday spread should persist even within the low-overnight-return subsample.

As Table 5 shows, even in the low-overnight-return subsample, the high-minus-low divergence quintiles exhibit a significant intraday spread of −1.50% per month, confirming that opinion divergence drives the intraday reversal independently of mechanical price pressure.[13] Conversely, the reversal is far more pronounced in the high-overnight-return group, consistent with the joint influence of belief divergence and price pressure: stocks more inflated at the open require more severe intraday corrections.

In addition, to shed light on whether our results are driven by transient price pressure or by belief-based overvaluation, we examine the time profile of price adjustment after the open. Prior studies on auction-related distortions document that a sizable fraction of the initial dislocation can revert within minutes following the start of continuous trading, with the correction often concentrated in the first 10–15 min (Hauser et al., 2022; Duong et al., 2025; Bogousslavsky and Muravyev, 2023). Motivated by these evidence, we extend the analysis to eight "half-hour" periods of continuous trading to assess whether the post-auction correction is largely front-loaded or continues to unfold over a longer horizon. Conceptually, mechanical price pressure implies a sharp and

[13] Furthermore, panel regressions (Appendix Table IV) reveal that controlling for contemporaneous overnight returns attenuates the $Divergence$ coefficient by 20% (from −0.0594 to −0.0473), yet its significance remains robust. This confirms that while opening price pressure partially contributes to the reversal, opinion divergence exerts a primary and independent influence on price formation during the call auction.

Table 5
Portfolio analysis of price pressure.

	High $R_t^{overnight}$	Low $R_t^{overnight}$
Low Divergence	2.31 (4.14)	6.92 (12.35)
2	1.84 (3.43)	5.91 (10.27)
3	1.50 (2.69)	5.74 (10.38)
4	0.28 (0.52)	5.45 (9.39)
High Divergence	−1.79 (−3.40)	5.42 (9.62)
High - Low	−4.10 (−13.13)	−1.50 (−4.54)

This table presents value-weighted (VW) overnight and intraday returns for portfolios sorted by opinion divergence. Each trading day, stocks are grouped by the median of daily overnight return (High $R_t^{overnight}$ and Low $R_t^{overnight}$). Within each subsample, stocks are sorted into quintiles by divergence of investor opinion. 'Difference' represents the return difference between the two subsamples. All returns are adjusted to a monthly frequency (daily returns × 21) and are expressed in percentages. The sample period is from January 4, 2010 to December 24, 2024. Newey–West adjusted t-statistics are reported in parentheses.

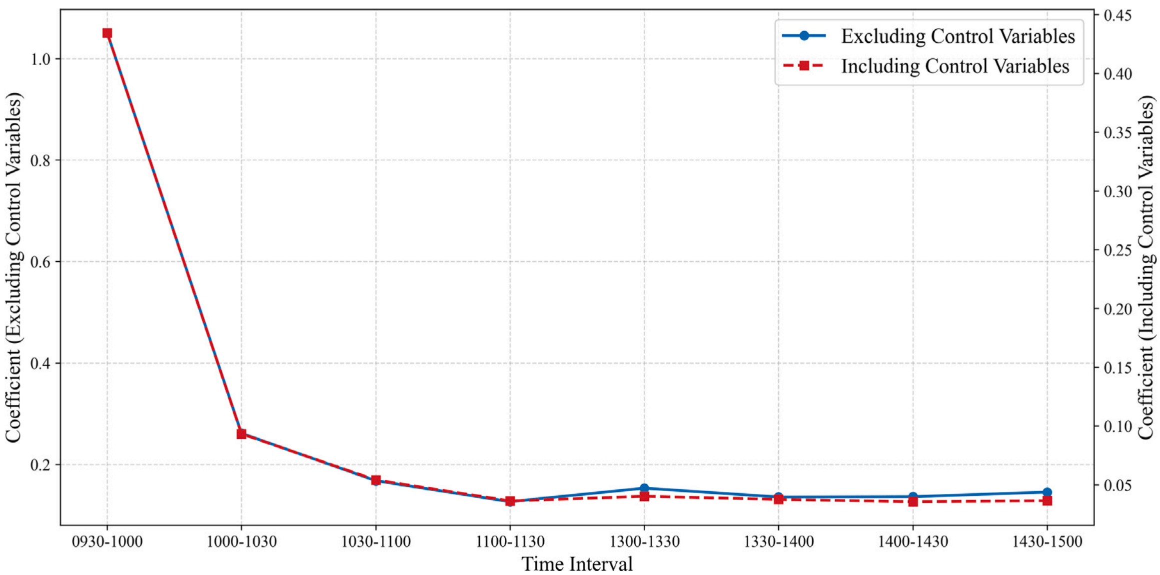

Fig. 3. The coefficient of $Divergence$
This figure illustrates the time-varying path of the $Divergence$ coefficient estimated from Eq. (5). The dependent variable is $RV_{i,t,interval}$, defined as the realized volatility of stock i over the $interval$ period on trading day t. We divide a complete trading day into eight "half-hour" periods: 9:30–10:00, 10:00–10:30, 10:30–11:00, 11:00–11:30, 13:00–13:30, 13:30–14:00, 14:00–14:30, 14:30–15:00. The key explanatory variable is $Divergence_{i,t}$, defined as the volume-weighted standard deviation of stock i's buy-order prices during the call auction from 9:20 to 9:25 (excluding 9:25) on day t, and divided by the previous day's closing price. The blue line shows the results without control variables, and the red line shows the results with control variables. The sample period is from January 4, 2010 to December 24, 2024.

short-lived adjustment, whereas disagreement-induced overvaluation is expected to dissipate more gradually. We therefore regress 30-minute realized volatility (RV) on $Divergence$ across intraday intervals to trace the dynamic association between disagreement and volatility, providing evidence on whether the divergence effect is transitory or persistent. The regression is shown below:

$$RV_{i,t,interval} = \alpha_t + \beta_1 \cdot Divergence_{i,t} + \lambda \cdot \boldsymbol{Controls} + \gamma_i + \sigma_t + \epsilon_{i,t} \quad (5)$$

where $RV_{i,t,interval}$ is the realized volatility of stock i over the $interval$ period on trading day t. We divide a complete trading day into eight "half-hour" periods: 9:30–10:00, 10:00–10:30, 10:30–11:00, 11:00–11:30, 13:00–13:30, 13:30–14:00, 14:00–14:30, 14:30–15:00.

Table 6
Robustness check: Closing call auction.

Panel A: Predict return								
	y = $R_{i,t}^{14:57-15:00}$				y = $R_{i,t+1}^{overnight}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Divergence_{i,t}^{14:57-15:00}$	0.0169 (27.03)	0.0148 (25.63)	0.0169 (27.03)	0.0148 (25.63)	−0.0498 (−19.73)	−0.0075 (−4.61)	−0.0498 (−19.72)	−0.0075 (−4.60)
Characteristics Variable	NO	YES	NO	YES	NO	YES	NO	YES
Events Variable	NO	NO	YES	YES	NO	NO	YES	YES
Stock FE	YES	YES	YES	YES	YES	YES	YES	YES
Trading Day FE	YES	YES	YES	YES	YES	YES	YES	YES
R^2	0.27%	11.55%	0.27%	11.55%	0.10%	13.13%	0.10%	13.13%

Panel B: Heterogeneity analysis								
	y = $R_{i,t}^{14:57-15:00}$				y = $R_{i,t+1}^{overnight}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Divergence_{i,t}^{14:57-15:00}$	0.0626 (8.93)	0.0189 (26.57)	0.0024 (1.20)	0.0012 (0.85)	−0.1436 (−5.49)	−0.0118 (−6.16)	0.0479 (11.34)	0.0044 (2.28)
$Divergence_{i,t}^{14:57-15:00} \times LogME_{i,t}$	−0.0072 (−6.92)				0.0204 (5.25)			
$Divergence_{i,t}^{14:57-15:00} \times \mathbf{I}_{i,t}^{MarginTrading}$		−0.0108 (−15.54)				0.0113 (4.74)		
$Divergence_{i,t}^{14:57-15:00} \times RES_{i,t}$			0.0669 (5.59)				−0.2964 (−12.08)	
$Divergence_{i,t}^{14:57-15:00} \times RV_{i,t}$				0.1419 (8.60)				−0.1230 (−5.68)
Characteristics Variable	YES	YES	YES	YES	YES	YES	YES	YES
Events Variable	YES	YES	YES	YES	YES	YES	YES	YES
Stock FE	YES	YES	YES	YES	YES	YES	YES	YES
Trading Day FE	YES	YES	YES	YES	YES	YES	YES	YES
R^2	11.56%	11.59%	11.61%	12.23%	13.14%	13.14%	13.17%	13.15%

This table reports the estimation results for the panel regression based on data from the closing call auction. The key explanatory variable is $Divergence_{i,t}^{14:57-15:00}$, defined as the volume-weighted standard deviation of stock i's buy-order prices during the closing call auction from 14:57 to 15:00 (excluding 15:00) on day t, and divided by the price at 14:57. The dependent variables are stock i's return over the 14:57–15:00 interval on day t and its overnight return on day $t+1$. Proxies for the strength of short-sale constraints include the log (base 10) of market capitalization ($LogME$), an indicator for margin-trading eligibility ($\mathbf{I}_{i,t-1}^{MarginTrading}$ equals 1 if eligible, 0 otherwise), relative effective spread (RES), and realized volatility (RV) on day t. Other control variables include stock i's return (R), ROE, and industry return ($R^{industry}$), the natural logarithm of one plus the total number of buy orders ($OrderNumber$), the total monetary value of these buy orders normalized by the stock's market capitalization on the previous day ($BuyPressure$), and indicators for whether earnings ($\mathbf{I}^{Earning}$), violation ($\mathbf{I}^{Violation}$), or M&A ($\mathbf{I}^{M\&A}$) announcements are released on day t (please refer to the Appendix for the definitions and construction of these variables). Standard errors are two-dimensionally clustered, and t-values are reported in parentheses. The sample period is from January 4, 2010 to December 24, 2024.

Fig. 3 shows the impact of $Divergence$ on RV peaking at 9:30–10:00 and stabilizing after 10:30. This indicates that while early volatility stems from both disagreement and price pressure, the persistent adjustment after 10:30 aligns with the smooth overvaluation-correction predicted by Miller (1977).[14]

This volatility trend mirrors the cumulative returns in Fig. 2. The long-short spread drops from 0.15% to 0.10% immediately after the open, followed by a monotonic decline throughout the day. Crucially, the post-10:30 reversal (Appendix Table VI) accounts for 64% of the total correction (2.24% of 3.48%) and persists across every 30-minute interval. This sustained underperformance confirms that the primary driver is a day-long correction of belief-driven overvaluation, rather than transient opening-period price pressure.

In summary, the "overvaluation-correction" pattern is primarily driven by opinion divergence. Despite data constraints regarding sell-side records, our battery of tests demonstrates that while price pressure contributes to the initial run-up, it is not the dominant driver. The robust predictive power of disagreement after isolating price pressure confirms its independent and significant role in call auction price formation.

6. Robustness check

To assess the robustness of our findings, we replicate the analysis using the closing call auction (14:57–15:00). This window operates under the same mechanism as the opening auction – orders can be submitted but not canceled, and prices are set at 15:00 – so we construct an analogous divergence measure and examine its relation to the closing return[15] and the next day's overnight

[14] Please see Appendix Table V for the regression results.

return. Table 6 shows that the closing-auction results closely mirror those from the open: opinion divergence is strongly positively associated with the same-day closing return and reverses at the next day's open, with stronger effects for stocks facing tighter short-sale constraints. This consistency across trading windows substantially strengthens the robustness of our main findings.

We further conduct robustness tests using alternative specifications: an alternative order collection window (9:15–9:20), an unweighted standard deviation of bid prices, two sub-periods (2010–2017 and 2018–2024), and winsorized order prices at the 0.5%, 1%, 2.5%, and 5% levels. Results remain qualitatively unchanged across all specifications, confirming that our conclusions are not contingent upon specific measurement choices, sample periods, or extreme observations (Appendix Tables VII and VIII).

7. Conclusion

7.1. Main findings

This study leverages the unique institutional design of the Chinese opening call auction to provide a granular examination of the relationship between investor opinion divergence and stock returns under short-sale frictions. By focusing on the non-cancellable order window, we effectively isolate the formation of belief disagreement from the subsequent price determination, thereby addressing the endogeneity and reverse-causality concerns that often plague low-frequency studies. Our empirical evidence reveals that high buy-side divergence leads to significant opening price run-ups, followed by a pronounced intraday reversal—a pattern that is notably amplified for stocks with tighter short-sale constraints. These results provide robust support for Miller (1977)'s overvaluation-correction hypothesis and offer new insights into how high-frequency disagreement-driven mispricing manifests and dissipates within a single trading day—a dimension that low-frequency studies have been unable to capture.

7.2. Implications and future research

Our findings carry practical implications for regulators, market designers, and traders. Regulators may use order-implied dispersion and auction activity to identify opening-price vulnerability, particularly for stocks with binding short-sale constraints. For exchanges, auction design parameters – such as order-book transparency, cancellation rules, and the transition to continuous trading – offer scope to improve price efficiency without undermining liquidity. For traders, buy-side divergence provides an ex ante signal of short-horizon dynamics useful for execution timing and intraday risk management.

Although this study leverages the unique temporal structure of the call auction to provide clear evidence for the pricing effects of disagreement, several avenues for future research remain. First, incorporating sell-side order data, when available, would enable a more comprehensive test of the (Miller, 1977) framework by directly characterizing pessimists' role in opening price formation. Second, cross-market comparisons could shed light on how institutional parameters – such as transparency, bidding rules, and investor structure – shape the aggregation of opinions and price discovery efficiency, yielding more universal evidence on market microstructure design.

CRediT authorship contribution statement

Yuntian Zhang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yongjie Zhang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Zhenao Guo:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation.

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Declaration of competing interest

We have no potential competing interests include employment, consultancies, stock ownership, honoraria, paid expert testimony, patent applications/registrations, and grants or other funding.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.frl.2026.109927.

[15] We calculate the return for each stock's closing price at 15:00 relative to its price at 14:57.

Data availability

The authors do not have permission to share data.

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